Review retrospective: *the Gaussian tail-bound foundation, and a third "`open Topology` is dead" over-flag*

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Abstract

A soundness review of `Laplace/OneD/TailBound.lean`: the elementary 1D Gaussian (Mill's-ratio) tail bounds and the FTC-2 first/second-moment closed forms that feed Laplace localisation. Result: **a clean fidelity pass with one dead-code removal** — four unused proof-locals. The review's most instructive moments were two *rejections*: the independent reviewer's recommendation to drop `open Topology` (rebuild-refuted — $\mathcal{N}$ is Topology-scoped, the third time this exact over-flag has appeared), and a linter-flagged "`congr 1` does nothing" that the seabed's own `ring`-under-`exp` caveat marks as plausibly load-bearing.

1 Setting

This file is the technical engine behind localisation: away from the minimiser, the Gibbs integrand $\exp(-tL)$ is exponentially suppressed, and the quantitative form of that suppression is a Gaussian tail bound. Its headline consumers are Localisation (which uses `gaussian_tail_bound_rescaled_Ioi` and the rescaled second-moment closed form with $b = \lambda t$) and `QuarticBoundedPrior`. It carries no named primer theorem, so the fidelity ground truth is the file's own (rich) docstrings checked against its statements, plus the downstream contract. It was selected as a never-reviewed foundation directly below an already-reviewed consumer.

2 What was reviewed

Nine public theorems and three private lemmas. The public surface:

- $\int_{(M,\infty)} x\, e^{-x^2/2} = e^{-M^2/2}$ and the Mill's ratio $\int_{(M,\infty)} e^{-x^2/2} \le e^{-M^2/2}/M$ $(M > 0)$;
- lower-tail = upper-tail by even symmetry, and the two-sided $\le 2e^{-M^2/2}/M$;
- $\int_{(M,\infty)} (x^2 - 1)e^{-x^2/2} = Me^{-M^2/2}$ and the second-moment decomposition $\int_{(M,\infty)} x^2 e^{-x^2/2} = Me^{-M^2/2} + \int_{(M,\infty)} e^{-x^2/2}$;
- the three $b > 0$ rescaled variants (first moment $e^{-bM^2/2}/b$; second moment; rescaled Mill's ratio $\le e^{-bM^2/2}/(bM)$), obtained by the substitution $u = x\sqrt{b}$.

The informal corpus was the file's docstrings, the Localisation review, the seabed `CLAUDE.md`, and a pre-check confirming the Mathlib homes of the imported Gaussian lemmas.

3 Fidelity / soundness findings

Sound; no mismatch. Each closed form was checked against its FTC-2 antiderivative ($-e^{-x^2/2}$ for the first moment, $-xe^{-x^2/2}$ for the (x^2-1) identity) and the $u = x\sqrt{b}$ rescaling; the constants, signs, the strictness of the $M > 0$ / $b > 0$ hypotheses, and the factor 2 in the two-sided bound all match. The reviewer independently reported no fidelity problems. A clean pass is itself the result here: the file that localisation rests on says exactly what it claims.

4 Simplifications applied

One: four dead proof-locals removed. Each rescaled theorem opens with `set s := Real.sqrt b with hs_def`, but the named defining equation `hs_def` is referenced nowhere — the `set` substitution does all the work. And in `gaussian_tail_bound_rescaled_Ioi` the local `hs_ne : s` $\neq$ `0` is unused (unlike the other two rescaled theorems, where it feeds `mul_div_cancel_right₀`); it was removed *only* there. Pure dead-code: rebuild green at 2694 jobs, the full-linter warning set byte-identical to baseline, and all twelve signatures byte-identical (the diff touches only intra-proof `set`/`have` lines). GPT affirmed.

5 Disagreements and open questions

The reviewer's recommendation to remove `open Topology` was **rejected by rebuild arbitration**: deleting it makes $\mathcal{N}$ an unknown identifier (five sites). The reviewer had attributed $\mathcal{N}$ to `Filter`, but the `nhds` notation is `Topology`-scoped. This is the *third* time the same "`open Topology` is dead" over-flag has been raised on a J-function/tail file and refuted the same way — the build, not the reviewer, is the arbiter of a load-bearing open. No open question; the open stays.

6 Lessons

- **A recurring reviewer blind spot is itself a datum.** "`open Topology` is dead" has now been wrong three times for the same reason ($\mathcal{N}$ is Topology-scoped, and a model reading proofs lumps it with `Filter`). The cheap guard is permanent: a one-line rebuild without the open. The reviewer's value is the fidelity pass and the dead-local catch, not its namespace bookkeeping.
- **Not every linter warning is a review edit.** The `unusedTactic` linter flags `congr 1; congr 1` at one site, but the seabed `CLAUDE.md` records that `ring` cannot unify under `exp` — and the goal there equates two `exp` atoms with arithmetically-equal but syntactically-distinct arguments. The `congr` chain is plausibly what lets the final `ring` see matching atoms; the reviewer did not flag it; touching a pre-existing warning outside the independent findings invites a thrash for no coverage gain. Left as a report-only observation.
- **"Dead local" generalises cleanly.** As with the `h`α`_ne` removal in the asymptotic-equivalence wrapper, a derived `have`/`set-with` binding has no signature consequence, so the only failure mode is an implicit tactic dependence — caught by the rebuild, confirmed by the `unusedVariables` linter's silence.

7 Follow-ups

- **Forward tide (structural):** extract two private helpers the reviewer correctly identified — (i) "from `IntegrableOn f (Ioi 0)` and `ContinuousOn f (Icc M 0)` conclude `IntegrableOn f (Ioi M)`", which collapses the ∼5 copies of the `by_cases hM` ∪-split, and (ii) the shared $\sqrt{b}$ algebra (`set s`, `hsq`, $(xs)^2 = bx^2$) across the rescaled family. Both add a declaration, so they are forward tides, not review edits.
- **Hygiene tide:** migrate the five `push_neg` occurrences to `push Not` (deprecated), and re-examine the line-300 `congr` chain with the LSP to settle whether `ring` alone closes it.
- The localisation sub-arc (`TailBound` → `Localisation`) is now reviewed end-to-end. Remaining laplace surface: the `TwoD/` families, `OneD/NearDegenerate`, `OneD/UniversalAsymptotics`, and the large `Multi/Covariance*` trio.